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Bisection - Method

OR

(Bolzano Method)

⇒ 1st step:

Intervals are given. If interval are not given then we assumed itself "2" intervals which give one value positive & one negative.

For Example

$$x_0 = 0 \text{ and } x_1 = 1$$

⇒ 2nd Step:

Put the intervals in given " $f(x)$ " function.

Also put the value of x_0 & x_1 which give values one positive one negative

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then

here $f(0) \cdot f(1) < 0$ So the roots
lies b/w 0 & 1. — (A)

⇒ 3rd step:

To Find x_2 use formula:

$$x_2 = \frac{x_0 + x_1}{2} \quad \text{--- (i)}$$

Then put the value of x_2 in
 $f(x)$ function.

After putting the value if function
 $f(x)$ give positive value then
choose the above roots which already
have negative value.

Otherwise

If $f(x)$ give negative value then choose
the above roots which already have
positive value.

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Then concluded the result
as we concluded in eq
(i).

⇒ 4th step:

To Find x_3 use formula:

$$x_3 = \frac{x_0 + x_2}{2} \quad \text{--- (ii)}$$

Then Pick the value of x_2
from (i) put in (ii).

After this put the value of
 x_3 in $f(x)$ function.

Now steps are continuous
which we discuss in
step 3.

(Bisection Method.)

Ex: solve $x^3 - 9x + 1$ for the root between $x = 2$ and $x = 4$ by bisection Method.

Solution;

Interval $(2, 4)$

$$f(x) = x^3 - 9x + 1 = 0$$

$$f(2) = (2)^3 - 9(2) + 1 = 8 - 18 + 1 = -9$$

$$f(4) = (4)^3 - 9(4) + 1 = 64 - 36 + 1 = 29$$

Here $f(2) f(4) < 0$ so Root lies between 2 and 4.

$$x_0 = 2, \quad x_1 = 4$$

$$x_2 = \frac{x_0 + x_1}{2} = \frac{2 + 4}{2} = 3$$

$$f(3) = (3)^3 - 9(3) + 1 = 27 - 27 + 1 = 1$$

Here $f(2) f(3) < 0$ so the Root lies between 2 and 3.

$$x_3 = \frac{x_0 + x_2}{2} = \frac{2 + 3}{2} = 2.5$$

$$f(2.5) = (2.5)^3 - 9(2.5) + 1 = -5.875 < 0$$

so the Root lies between 2.5 & 3 as

$$f(2.5) f(3) < 0$$

$$x_4 = \frac{2.5 + 3}{2} = 2.75$$

Ex#02; Carry out Five iterations for the function $f(x) = 2x \cos(2x) - (x+1)^2$ for the interval $(-1, 0)$

$$f(x) = 2x \cos(2x) - (x+1)^2$$

$$f(-1) = 2(-1) \cos(-2) - (-1+1)^2 \\ = -2(-0.4161) = 0.8322 > 0$$

$$f(0) = 2(0) \cos(0) - (0+1)^2 = -1 < 0$$

So the roots lie between 0 and (-1) as

$$f(0)f(-1) < 0$$

$$x_2 = \frac{x_0 + x_1}{2} = \frac{0 - 1}{2} = -0.5$$

$$f(-0.5) = 2(-0.5) \cos(-1) - (-0.5+1)^2 = -0.7903 < 0$$

Roots lie between $f(-1)f(-0.5) < 0$

$$x_3 = \frac{x_0 + x_2}{2} = \frac{-0.5 - 1}{2} = -0.75$$

Same Find x_4 & x_5

Example; Carry out first five iterations

$$f(x) = x \cos x - 2x^2 + 3x - 1$$

$$1.2 \leq x \leq 1.3$$

$$f(1.2) = (1.2) \cos(1.2) - 2(1.2)^2 + 3(1.2) - 1 \\ = 0.154829$$

$$f(1.3) = (1.3) \cos(1.3) - 2(1.3)^2 + 3(1.3) - 1 \\ = -0.13225$$

So Roots lie between 1.2 & 1.3

$$f(1.2)f(1.3) < 0$$

$$x_0 = 1.2, x_1 = 1.3$$

$$x_2 = \frac{x_0 + x_1}{2} = \frac{1.2 + 1.3}{2} = 1.25$$

$$\textcircled{2} f(1.25) = (1.25) \cos(1.25) - 2(1.25)^2 + 3(1.25) - 1$$

$$= 0.01917 > 0$$

$f(1.25)f(1.3) < 0$ so the Roots lie between 1.25 and 1.3

$$x_3 = \frac{1.25 + 1.3}{2} = \cancel{1.275} \quad 1.275$$

$$f(-0.025) = (-0.025) \cos(-0.025) - 2(-0.025)^2 + 3(-0.025) - 1$$

$$= -0.0545 < 0$$

$f(1.25) \& f(1.275) < 0$ so the Root lie between both

$$x_4 = \frac{1.25 + 1.275}{2} = 1.2625 > 0$$

$$f(1.2625) = \text{put in function}$$

$$= -0.0545 < 0$$

$f(1.25)f(1.275) < 0$ so Root lie between both.

$$x_5 = \frac{1.25 + 1.2625}{2} = 1.25625$$

$$f(1.25625) =$$

$$= 0.00108 > 0$$

$f(1.25625)f(1.2625) < 0$ so Roots lie between both;

$$x_6 = \frac{1.25625 + 1.2625}{2} = 1.259375.$$

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— Regula-Falsi Method

OR

Method of False Position

Step 1:

Same as bisection method.

Step 2:

Here put the value of x_1 & x_2 in formula:

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

when we put the interval x_1 & x_2 in $f(x)$ function we find $f(x_1)$ & $f(x_2)$.

Next procedure is same as Bisection method.

★ Regula-Falsi Method (Method of false position)

Formula;

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

Ex:- use Regula Falsi Method to compute a Real Root of the equation

$x^3 - 9x + 1 = 0$ if Roots lie between 2 & 4.

Sol; $x_1 = 2, x_2 = 4$

	$f(x_1)$
$f(2) = 2^3 - 9(2) + 1 = -9$	
$f(4) = 4^3 - 9(4) + 1 = 29$	$f(x_2)$

For $n = 2$

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

$$x_{2+1} = x_2 - \frac{x_2 - x_{2-1}}{f(x_2) - f(x_{2-1})} f(x_2)$$

$$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2)$$

$$x_3 = 4 - \frac{4 - 2}{29 - (-9)} (29) = 4 - \frac{2(29)}{38} = 2.4736$$

$$f(x_3) = f(2.4736) = 2.4736^3 - 9(2.4736) + 1 = -6.12644$$

$f(x_2) f(x_3) < 0$ Roots lie between x_2 & x_3

③ Put $n=3$

$$x_4 = x_3 - \frac{x_3 - x_2}{f(x_3) - f(x_2)} f(x_3)$$

$$= 2.4736 - \frac{2.4736 - 4}{-6.12644 - 29} (-6.12644)$$

$$x_4 = 2.73989$$

$$f(x_4) = f(2.73989) = 2.73989^3 - 9(2.73989) + 1 = -3.090707$$

Since $f(x_4)f(x_2) < 0$ so Roots lie between x_4 & x_2 .

put; Next approximation is obtained from,
 $n=4$

$$x_5 = x_4 - \frac{x_4 - x_3}{f(x_4) - f(x_3)} f(x_4)$$

$$= 2.73989 - \frac{2.73989 - 2.4736}{-3.090707 + 6.12644} (-3.090707)$$

$$x_5 = 3.01100$$

$$f(x_5) = (3.01100)^3 - 9(3.01100) + 1 = \underline{1.19909}$$

Difference Between Bisection & Regula Falsi

In Bisection we have to choose Interval when $f(x_0)f(x_1) < 0$. And Important difference is that in Bisection we use last Approximation value for Finding the Interval of Next approximation.

But in Regula Falsi we have to use every Time 1st $f(x)$ in order to find the Next approximation.

And Formula for Finding Iterations in Regula Falsi Method is

$$f(x_{n+1}) = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

* In Regula Falsi when cos, sin and e functions includes in Given $f(x)$ we have to take x_0 and x_1 as Starting Interval.

and when polynomial and Quadratic equation is given we have to take x_1 and x_2 as a Starting Interval.

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Method of Iteration

Step 1:

First $f(x)=0$ is rewrite in the form

$$x = \phi(x)$$

Step 2:

let two intervals

$f(0)$ & $f(1)$ ^{Put in $x = \phi(x)$} which gives

one value > 0 and one < 0 .

So, Root lie b/w 0 & 1.

Step 3:

Take the derivate of function

$x = \phi(x)$ which become $\phi'(x)$.

Put the value of $f(0)$ & $f(1)$ in

$\phi'(x)$ which give value > 0 & < 0 .

* Method of Iteration

Method of Iteration can be applied to find a Real Root of the equation $f(x)=0$ by rewriting the system in the form.

$$x = \phi(x)$$

Example;

$$f(x) = \cos x - 2x + 3 = 0$$

It can be rewritten as; $x = \frac{1}{2}(\cos x + 3) = \phi(x)$

$$\phi(x) = \frac{1}{2}(\cos x + 3) \rightarrow \textcircled{1}$$

$$f(x) = \cos x - 2x + 3 = 0$$

$$f(1) = \cos(1) - 2(1) + 3 = 0 = 1.54030 > 0$$

$$f(2) = \cos(2) - 2(2) + 3 = 0 = -0.04161 < 0$$

So Roots lies between 1 & 2

$$\phi'(x) = -\frac{1}{2}(\sin x)$$

$$\phi'(1) = -\frac{1}{2} \sin(1) = -0.4207$$

$$\phi'(2) = -\frac{1}{2} \sin(2) = -0.4207$$

$\phi'(1) \& \phi'(2) < 1$ so the Method of iteration can be applied;

$$\text{let } x_0 = 1.5$$

$$x_1 = \frac{1}{2}(\cos x_0 + 3) = \frac{1}{2}(\cos(1.5) + 3) = 1.9998$$

$$x_2 = \frac{1}{2}(\cos x_1 + 3) = \frac{1}{2}(\cos(1.9998) + 3) = 1.9996$$

$$x_3 = \frac{1}{2}(\cos x_2 + 3) = \frac{1}{2}(\cos(1.9996) + 3) = 1.99969$$

So, this is the Required Roots upto 5 decimal places.

★ Newton-Raphson Method

FORMULA USED FOR NEWTON RAPHSON METHOD;

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \text{for } n = 0, 1, 2, 3 \dots$$

★ Newton's Raphson Method is also referred to as the "Method of tangents".

Example:-

Find the real Root of the equation $x^3 - x - 1 = 0$ using Newton's-Raphson Method, correct to Four decimal places.

Solution:-

$$f(x) = x^3 - x - 1$$

$$f(1) = (1)^3 - (1) - 1 = -1 < 0$$

$$f(2) = (2)^3 - (2) - 1 = 5 > 0$$

So the Roots lie between 1 & 2

$$\text{Here } f'(x) = 3x^2 - 1$$

$$f'(1) = 3(1)^2 - 1 = 2$$

$$f'(2) = 3(2)^2 - 1 = 11$$

$$\text{So, } x_0 = 1$$

The second approximation is computed using Newton-Raphson Method as;

$$N=0 \\ x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 - \frac{5}{11} = 1.54545$$

$$f(1.54545) = (1.54545)^3 - (1.54545) - 1 = 1.14576$$

$$f'(x) = 3x^2 - 1 = 3(1.54545)^2 - 1 = 6.16487$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1.54545 - \frac{1.14576}{6.16487} = 1.35961$$

$$f(1.35961) = (1.35961)^3 - (1.35961) - 1 = 1.14576$$

$$f'(1.35961) = 3(1.35961)^2 - 1 = 6.16487$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.35961 - \frac{1.14576}{6.16487}$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = \underline{\hspace{2cm}}$$

$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)} = \underline{1.3247}$$

Example:-

Set up Newton's Scheme of iteration for Finding the square root of a positive Number N.

Solution:-

The Square root of N can be carried out as a root of the equation;

$$x^2 - N = 0$$

$$\text{Let, } f(x) = x^2 - N$$

By Newton's Method, we have;

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x) = x^2 - N ; f'(x) = 2x$$

Therefore;

$$x_{n+1} = x_n - \frac{x_n^2 - N}{2x_n} = \frac{1}{2} \left(x_n + \frac{N}{x_n} \right)$$

* Newton's Method.

- * In Newton's Method we have to put Min value in $f(x)$ like $(0, 1, 2, -1, -2)$ etc, we need Answer from Negative to positive or positive to negative.
- * If Intervals are give use them in $f(x)$.
- * Then we have to take derivative of given Function, and put the same values in $f'(x)$.

- * Now The ~~value~~ solution is greater in both $f(x)$ and $f'(x)$ use that value as a First approximation; $x_0 = \underline{\hspace{2cm}}$.

And then Continue Finding

like;
Iterations;

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

- * Put values of x_1 & $x_2 \dots x_n$ in both Functions $f(x)$ and $f'(x)$ for Finding Next Iteration.

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Secant Method

1st step:

We let the two intervals and then put in $f(x)$ function.

x_0 & x_1

2nd step:

We do not need to check the interval that are " > 0 " & " < 0 ". And also not to check the roots lie b/w the successive approximation in secant method.

3rd step:

To Find the value of x_2 use formula: -

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For $n=1$

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

For $n=2, 3, \dots$ use thisformula to find the
next all iterations.

Secant Method

The Secant method is Modified Form of Newton-Raphson method.

"we replace $f'(x_n)$ by the difference ratio"

* General form of Secant Method;

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

Example;

Do three iterations of Secant Method to Find the root of;

$$f(x) = x^3 - 3x + 1 = 0$$

Taking; $x_0 = 1, x_1 = 0.5$

* (we can also take $(0, 1, -1, 2)$)

But we have to choose Min value

$$f(x_0) = f(1) = (1)^3 - 3(1) + 1 = -1$$

$$f(x_1) = f(0.5) = (0.5)^3 - 3(0.5) + 1 =$$

for $n = 1$

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$= \frac{(1)(-0.375) - (0.5)(-1)}{-0.375 - (-1)} = 0.2$$

For $n=2$

$$f(x_2) = f(0.2) = (0.2)^3 - 3(0.2) + 1 = 0.408$$

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)}$$
$$= \frac{(0.5)(0.408) - (0.2)(-0.375)}{0.408 - (-0.375)} = 0.3563$$

For $n=3$

$$f(x_3) = f(0.3563) = (0.3563)^3 - 3(0.3563) + 1$$
$$= -0.02367$$

$$x_4 = \frac{x_2 f(x_3) - x_3 f(x_2)}{f(x_3) - f(x_2)}$$
$$= \frac{(0.2)(-0.02367) - 0.3563(0.408)}{(-0.02367) - (0.408)}$$

$$x_4 = 0.3473$$

Same Find:-

$$x_5 = -0.0000096$$

Ans At the end we have to find difference absolute value of last two iterations;

Like;

$$|x_5 - x_4| = 0.00004$$

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Muller Method

1st step:

It requires three steps.

x_0 , x_1 and x_2 .

It does not require the derivative of function.

2nd step:

2nd we put the value of x_0 , x_1 and x_2 in function given.

3rd step:

In Muller method we use formula:-

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Muller's Method Formula:

$$X = X_0 - \frac{2c}{b - \sqrt{b^2 - 4ac}}$$

here we need the value of a, b & c which we obtained by using formula:-

$$c = f_0$$

$$b = \frac{f_1 - f_0 - ah_1^2}{h_1}$$

$$a = \frac{h_1 f_2 - (h_1 + h_2) f_0 + h_2 f_1}{h_1 h_2 (h_1 + h_2)}$$

By using these formula's we find 1st iteration.

For 2nd iteration

$$h_1 = \frac{x_1 - x_0}{\text{initial } x_0}$$

$$h_2 = x_0 - x_2$$

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4th step:

For 2nd iteration

Mullar's method کے فارمولا سے جو

x_0 کی value نکالی ہے وہ x_1

لے گئے اور x_1 کی value

لے گئے جو question میں x_0 کی دی

گئی ہوگی اور x_1 کی وہی لے گئے

جو question میں دی گئی ہو۔

In this way we find
2nd iteration.

we also find the next
iteration by using 2nd iteration
concept.

★ Muller's Method

- ⇒ Muller's Method is a generalization of the Secant Method.
- ⇒ It does not require the derivative of the Function. It is an Iterative Method that requires three starting points.
 x_0, x_1, x_2 .

Example;

Do two Iterations of Muller's Method to solve $x^3 - 3x + 1 = 0$ starting with;
 $x_0 = 0.5, x_2 = 0, x_1 = 1$

Solution;

$$f(x_0) = f(0.5) = (0.5)^3 - 3(0.5) + 1 = -0.375$$

$$f(x_1) = f(1) = (1)^3 - 3(1) + 1 = -1$$

$$f(x_2) = f(0) = 0 - 3 \times 0 + 1 = 1$$

$$c = f(x_0) = -0.375$$

$$h_1 = x_1 - x_2 = 0.5$$

$$h_2 = x_0 - x_1 = 0.5$$

$$a = \frac{h_2 f_1 - (h_1 + h_2) f_0 + h_1 f_2}{h_1 h_2 (h_1 + h_2)}$$

$$= \frac{(0.5)(-1) - (-0.375) + (0.5)}{0.25}$$

$$a = 1.5$$

$$b = \frac{f_1 - f_0 - ah_1^2}{h_1} = -2$$

∴ Formula for Finding Iterations in Muller's Method;

$$x = x_0 - \frac{2c}{+b - \sqrt{b^2 - 4ac}}$$

$$= 0.5 - \frac{2(-0.375)}{-2 - \sqrt{4 + 4(1.5)(0.375)}}$$

$$= 0.5 - \frac{-0.75}{-2 - \sqrt{4 + 2.25}} = 0.3333 < 0.5$$

For 2nd Iteration;

$$x_2 = 0, \quad x_0 = 0.3333, \quad x_1 = 0.5$$

$$h_1 = x_1 - x_0 = 0.16667$$

$$h_2 = x_0 - x_2 = 0.3333$$

$$c = f_0 = 0.3333$$

$$f_0 = f(x_0) = 0.037046$$

$$f_1 = f(x_1) = -0.375$$

$$f_2 = f(x_2) = 1$$

$$a = \frac{h_2 f_1 - (h_1 + h_2) f_0 + h_1 f_2}{h_1 h_2 (h_1 + h_2)} = \frac{0.023148}{0.027778}$$

$$a = 0.8333$$

$$b = \frac{f_1 - f_0 - ah_1^2}{h_1} = -2.6$$

$$x = x_0 - \frac{2c}{b \pm \sqrt{b^2 - 4ac}}$$

$$= 0.333333 - \frac{0.074092}{-5.2236}$$

$$= 0.3475 > 0.3333 = x_0$$

For 3rd,

$$x_2 = 0.33333, \quad x_0 = 0.3475, \quad x_1 = 0.5$$

Solution Of Linear

System of Equations

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Gaussian Elimination Method

1st step:

Write the given system of equation in form

$$AX = B$$

Then

augmented matrix is

$$[A \mid B]$$

Step 2:

$$[A \mid B]$$

Making A reduce to upper triangular form

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Step 3:

Its reduce by using elementary transformation.

Elementary transformation is applied according to given system of the equation.

Step 4:

After applied elementary transformation upper triangular system is solved by using back substitution process.

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Example

$$2x + 3y - z = 5$$

$$4x + 4y - 3z = 3$$

$$-2x + 3y - z = 1$$

Sol:

step 1

$$\begin{bmatrix} 2 & 3 & -1 \\ 4 & 4 & -3 \\ -2 & 3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix}$$

step 2

$$\left[\begin{array}{ccc|c} 2 & 3 & -1 & 5 \\ 4 & 4 & -3 & 3 \\ 2 & 3 & -1 & 1 \end{array} \right]$$

step 3

Reduce to upper triangular form

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Divided the 1st equation by 2 and the subtract equation (multiply by 4 and -2) from 2nd and 3rd equation. Thus, we eliminate x from 2nd and 3rd equation.

The new resulting equation is

$$x + 3\frac{1}{2}y - \frac{1}{2}z = 5\frac{1}{2}$$

$$-2y - z = -7$$

$$6y - 2z = 6$$

we divide 2nd equation by "-2" and eliminate y from 3rd equation.

Thus new equation is

$$x + 3\frac{1}{2}y - \frac{1}{2}z = 5\frac{1}{2}$$

$$y + \frac{1}{2}z = 7$$

$$-5z = -15$$

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4th step

By Backward Substitution:

$$\boxed{z = 3}$$

Put value of "z" 2nd eq

$$y + \frac{3}{2} = \frac{7}{2} \Rightarrow y = \frac{7}{2} - \frac{3}{2}$$

$$\boxed{y = \frac{4}{2} = 2}$$

Put value of "y" and "z" in 1st eq

$$x + \frac{3}{2} \times z - \frac{3}{2} = \frac{5}{2}$$

$$x + 3 - \frac{3}{2} = \frac{5}{2}$$

$$x = \frac{5}{2} + \frac{3}{2} - 3 = \frac{5+3-6}{2}$$

$$x = \frac{2}{2}$$

$$\boxed{x = 1}$$

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Gauss-Jordan Elimination Method

Step 1:

The elements above and below the diagonal are made zero.

Step 2:

A equivalent diagonal form using elementary transformation.

After this solution of resulting diagonal system obtained.

Example:

$$x + 2y + z = 8$$

$$2x + 3y + 4z = 20$$

$$4x + 3y + 2z = 16$$

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Sol:

In matrix notation

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 4 \\ 4 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 20 \\ 16 \end{bmatrix}$$

Now

$(-2)R_1 + R_2$ & $(-4)R_1 + R_3$ then

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 2 \\ 0 & -5 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \\ -16 \end{bmatrix}$$

eliminate "y" from 1st & 3rd row

using 2nd row

$$\begin{bmatrix} 1 & 0 & 5 \\ 0 & -1 & 2 \\ 0 & 0 & -12 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ 4 \\ -36 \end{bmatrix}$$

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eliminat "z" from 1st and 2nd row
using Normalized 3rd row

$$\begin{bmatrix} 1 & 0 & 5 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ 4 \\ 3 \end{bmatrix}$$

eliminat 'z' from above equation

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$$

Solution is

$$x = 1$$

$$y = 2$$

$$z = 3$$

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$$[L][U] = [A]$$

$$\begin{bmatrix} a & 0 & 0 \\ d & b & 0 \\ e & f & c \end{bmatrix} \begin{bmatrix} 1 & g & h \\ 0 & 1 & i \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & -2 & 1 \\ 7 & 1 & -5 \\ 3 & 7 & 4 \end{bmatrix}$$

$$\begin{bmatrix} a & ag & ah \\ d & dg+bi & dh+bi \\ e & eg+f & eh+fi+c \end{bmatrix} = \begin{bmatrix} 5 & -2 & 1 \\ 7 & 1 & -5 \\ 3 & 7 & 4 \end{bmatrix}$$

By compare

$$a=5, ag=-2 \Rightarrow 5g=-2 \Rightarrow g = -2/5$$

$$ah=1 \Rightarrow 5h=1 \Rightarrow h = 1/5, d=7$$

 $dg+bi=1 \Rightarrow$ By putting values we

$$\text{obtained } dg + b = \frac{19}{5}, dh+bi = -5$$

By putting values

$$e=3$$

$$eg+f=3$$

by putting
value

$$f = \frac{41}{5}$$

$$eh+fi+c=4$$

By putting
value:-

$$c = \frac{327}{19}$$

$$i = -\frac{32}{19}$$

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Put all value in $[L]$ & $[U]$

$$[L][U][X] = [B]$$

$$\begin{bmatrix} 5 & 0 & 0 \\ 7 & 19/5 & 0 \\ 3 & 41/5 & 327/19 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 10 \end{bmatrix}$$

step 3^{By} substituting

$$[L][Y] = B$$

$$\begin{bmatrix} 5 & 0 & 0 \\ 7 & 19/5 & 0 \\ 3 & 41/5 & 327/19 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 10 \end{bmatrix}$$

By back ward substitution:

$$5y_1 = 4 \Rightarrow y_1 = \frac{4}{5}$$

$$5y_1 + \frac{19}{5}y_2 = 8 \Rightarrow y_2 = \frac{12}{19}$$

$$3y_1 + \frac{41}{5}y_2 + \frac{327}{19}y_3 = 10 \Rightarrow y_3 = \frac{46}{327}$$

step 4:

$$[U][X] = Y$$

$$\begin{bmatrix} 1 & -2/5 & 1/5 \\ 0 & 1 & -32/19 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4/5 \\ 12/19 \\ 46/327 \end{bmatrix}$$

$$x_1 - \frac{2}{5}x_2 + \frac{1}{5}x_3 = \frac{4}{5}$$

$$x_1 = \left(\frac{2}{5}\right)\left(\frac{12}{19}\right) + \left(\frac{1}{5}\right)\left(\frac{46}{327}\right) = \frac{4}{5}$$

$$x_1 = \frac{366}{327}$$

$$x_2 - \frac{32}{19}x_3 = \frac{12}{19}$$

$$x_2 = \left(\frac{32}{19}\right)\left(\frac{46}{327}\right) = \frac{12}{19}$$

$$x_2 = \frac{284}{327}$$

$$x_3 = \frac{46}{327}$$

$$x_1 = \frac{366}{327}, x_2 = \frac{284}{327}$$

$$x_3 = \frac{46}{327}$$

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Jacobi's Method of Matrix Inversion

This method understand by example.

$$8x - 3y + 2z = 20$$

$$4x + 11y - z = 33$$

$$6x + 3y + 12z = 35$$

solve

First we arrange the system is diagonally dominant. and

let the $x = \frac{1}{8} [20 + 3y - 2z]$

Right side

of equation $y = \frac{1}{11} [33 - 4x + z]$

$x = x_0$

$y = y_0$

$z = z_0$

$z = \frac{1}{12} [35 - 6x - 3y]$

Day:

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step 2:

For 1st iteration

we start with an initial
approximation $x_0 = y_0 = z_0 = 0$

Put in 1st step we obtained

$$x_1 = \frac{1}{8} [20 + 3(0) - 2(0)] = 2.5$$

$$y_1 = \frac{1}{11} [33 - 4(0) + 0] = 3$$

$$z_1 = \frac{1}{12} [35 - 6(0) - 3(0)] = 2.9167$$

step 3:

For 2nd iteration

$$x_2 = \frac{1}{8} [20 + 3y_1 - 2z_1]$$

$$y_2 = \frac{1}{11} [33 - 4x_1 + z_1]$$

$$z_2 = \frac{1}{12} [35 - 6x_1 - 3y_1]$$

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Put value of x_1, y_1 and z_1 in
Right side of equation, we
obtained the values of
2nd iteration.

Similarly All the next
iterations is find by
understanding this concept
of iteration.

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Gauss-Seidel Iteration Method

$$10x - 2y - z - w = 3$$

$$-2x + 10y - z - w = 15$$

$$-x - y + 10z - 2w = 27$$

$$-x - y - 2z + 10w = -9$$

Sol:

$$x = \frac{1}{10} [3 + 2y + z + w]$$

$$y = \frac{1}{10} [15 + 2x + z + w]$$

$$z = \frac{1}{10} [27 + x + y + 2w]$$

$$w = \frac{1}{10} [-9 + x + y + 2z]$$

Date:

Day:

step 2:

1st iteration

[Put $(y=z=w=0)$ on R.H.S of eq
in step 1]

$$x_1 = 0.3$$

[Put $(x=0.3, z=w=0)$ on R.H.S
of eq in step 1]

$$y_1 = 1.56$$

[Put $x=0.3, y=1.56$ and $w=0$ in
R.H.S of eq in step 1]

$$z_1 = 2.886$$

[Put $x=0.3, y=1.56$ & $z=2.886$ in
R.H.S of eq in step 1]

$$w_1 = -0.1368$$

Day:

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step 3:

For 2nd iteration

Pull all values of x_1, y_1, z_1, w_1 to find 2nd iteration

$$x_2 = \frac{1}{10} [3 + 2y_1 + z_1 + w_1]$$

$$y_2 = \frac{1}{10} [15 + 2x_1 + z_1 + w_1]$$

$$z_2 = \frac{1}{10} [27 + x_1 + y_1 + 2w_1]$$

$$w_2 = \frac{1}{10} [-9 + x_1 + y_1 + 2z_1]$$

step 4:

For next iterations

follow the step 3.

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— d Eigen value Problems of Power Method —

step 1:

Choose the initial vector such that largest element is unity.

initial vector اس کا مطلب ہے کہ ہم

کہ $[v^{(0)}]$ سے ظاہر کرتے ہیں اگر سوال میں

initial vector نہ دیا ہو تو $v^{(0)} = (1, 1, 1)$

OR $v^{(0)} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ ہم نے یہی لپٹا ہے ہمیشہ

یہ rule کا Power method ہے

step 2:

$v^{(0)}$ is pre-multiplied by the matrix $[A]$

ی > equation ہے اس کے معنی

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گنتی ہو گئی اسے $[A]$ کا نام دے کر $V^{(0)}$ سے $pre-multiplied$ کریں گے

$pre-multiplied$ کا مطلب یہ کہ $[A]$ سے $V^{(0)}$ کو $System (multiply)$ کرنا ہے

$$U^{(1)} = [A] V^{(0)}$$

Step 3:

In step 2
 $U^{(1)}$ (resultant vector) کو
 ظاہر کر لے گا۔

The resultant vector again
 normalized.

The result look at this
 Point is

$$U^{(k)} = [A] V^{(k-1)} = q_k V^{(k)}$$

Day:

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step 4:

In step 3

 q_k denote desire largest value $v^{(k)}$ corresponding eigenvector.

At the end, Process of iteration continued.

 q_k کی value کو μ_1 (λ)

سے replace کریں گے، مقام

iteration کرنے سے بعد اور

 $v^{(k)}$ کی value کو μ_1 (X)

سے replace کریں گے

Now, we do example of this method.

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Example

$$[A] = \begin{bmatrix} 2 & 3 & 2 \\ 4 & 3 & 5 \\ 3 & 2 & 9 \end{bmatrix}$$

1st step:

choose initial vector

$$v^{(0)} = (1, 1, 1)^T$$

Compute 1st iteration

$$u^{(1)} = [A][v^{(0)}] = \begin{bmatrix} 2 & 3 & 2 \\ 4 & 3 & 5 \\ 3 & 2 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 12 \\ 14 \end{bmatrix}$$

Now, normalize the resultant vector

$$u^{(1)} = 14 \begin{bmatrix} 1/2 \\ 6/7 \\ 1 \end{bmatrix}$$

$$v^{(1)} = \frac{1}{14} u^{(1)}$$

step 2

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Relaxation Method

step 1:

System of equation دیا لیا ہوگا۔ تمام equation کو R.H.S پر لے جائیں گے اور L.H.S والی کو R_x, R_y و R_z سے denote کریں گے

step 2:

1st eq میں ہم دیکھ گئے کہ جو ("x") کی value ہے وہ (y, z) کی نسبت بڑی ہو۔ اسی طرح 2nd eq میں ("y") کی بڑی ہو (x, z) کے مقابلے میں۔ 3rd eq میں ("z") میں کی بڑی ہو (x, y) کے مقابلے میں۔
Otherwise

آخر equation میں ایسا نہ ہو تو ہم نے (1st eq) کو اس eq سے replace کرنا ہے جس میں (x) کی value بڑی ہو۔ اور دوسری میں (y) کی اور تیسری میں (z) کی۔

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3rd step

طریقہ کے پہلے (Residuals increment) کا table بنانا ہے۔

Residuals increment	δR_x	δR_y	δR_z
δx			
δy			
δz			

اس میں پانچ value لکھنی ہے جو 1st step

میں P.H.S پر x, y, z کی دی گئی ہو۔

1st line میں 1st eq کی اسی طرح دوسری

میں 2nd eq کی اور 3rd line میں 3rd eq

کی اور یہ (Horizontal values) لکھنی پڑے گا۔

step 4:

اس کے بعد Relaxation Table بنانا ہے۔

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2nd iteration

$$U^{(1)} = [A][V^{(1)}] = \begin{bmatrix} 2 & 3 & 2 \\ 4 & 3 & 5 \\ 3 & 2 & 9 \end{bmatrix} \begin{bmatrix} 1/2 \\ 6/7 \\ 1 \end{bmatrix} = \begin{bmatrix} 39/7 \\ 67/7 \\ 17/14 \end{bmatrix}$$

$$U^{(2)} = 12.2143 \begin{bmatrix} 0.45614 \\ 0.78362 \\ 12.2143 \end{bmatrix} = 4.23412$$

$$U^{(2)} = \rho_{(2)} V^{(2)}$$

step 3:

Continue this process to Follow the steps to find all next iteration.

Jacobies Method for finding Eigen values and eigenvectors
I have added at the end of This file

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Example:

$$9x - y + 2z = 9$$

$$x + 10y - 2z = 15$$

$$2x - 2y - 13z = -17$$

$$9x - y + 2z = 9 \quad \text{--- (i)}$$

$$x + 10y - 2z = 15 \quad \text{--- (ii)}$$

$$2x - 2y - 13z = -17 \quad \text{--- (iii)}$$

$$R_x = 9 - 9x + y - 2z$$

$$R_y = 15 - x - 10y + 2z$$

$$R_z = -17 + 2x - 2y + 13z$$

Residual table or/operation table

number	R_x	R_y	R_z
δx	-9	-1	-2
δy	1	-10	+2
δz	-2	+2	13

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Relaxation table

	x	y	z	R_x	R_y	R_z
1.	0	0	0	9	15	-17
2.	0	0	1.3	6.4	17.6	-0.1
3.	0	1.76	1.3	8.16	0	3.42
4.	0.906	1.76	1.3	0.006	-0.906	1.608

اس کا طریقہ یہ ہے کہ سب سے پہلے $x=y=z=0$ لیتا ہے جو R_x, R_y, R_z میں Put کرنا ہے۔ پھر جو values آئے گی اس میں سب سے بڑی value لیتی ہے جیسا کہ اوپر Table میں R_z کی value سب سے بڑی ہیں۔ پھر ہم وہی value کو operation table میں (8z) میں دیکھنا ہے اور اگر وہ (+ive) ہے تو (-ive) لے گے اگر (-ive) ہے تو (+ive) لے گے۔ اور پھر بڑی value کو "z" والی value یہ تقسیم کرنا ہے۔ البتہ $(-17/1.3 = -13.07)$

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$$x = y = 0$$

پھر (2nd approximation)

جس کے $z = 1.3$ اب ان values کے R_x, R_y, R_z والی eq میں R_x کرنا ہے۔ پھر تمام

ولس میں Step کی طرح جاری رکھنا

ہے یہ Table کے اقدار سے ملے ہیں

کرنا ہے next all approxi

اور یہ تب تک جاری رکھنا ہے جب تک

ملکیں (0) سے قریب ترین value ہے

مل جائیں R_x, R_y, R_z کی

THE END

(Jacobi's Method)

(For Finding Eigen values)

Procedure;

① سب سے پہلے جو میٹرک دیا گیا ہوگا اس میٹرک کی سب سے بڑی دو off-diagonal ایلیمنٹ کو دیکھنا ہوتا ہے۔

② اگر off-diagonal میں بڑی ویلیو پہلی Row اور دوسرے کالم میں ہوگی تو اسکو ہم (a_{12}) کہہ سکتے ہیں۔ ہمیشہ دو ویلیوز ایک جیسی ہوتی ہیں off-diagonal۔

③ اگر off-diagonal میں بڑی ویلیو دوسری Row اور دوسرے کالم میں ہوگی اسکو (a_{22}) کہہ سکتے ہیں۔

Tips for using formula;

$$\tan 2\theta = \frac{2a_{ij}}{a_{ii} - a_{jj}}$$

اگر (a_{13}) ہو تو فرم کر لیں۔ $i=1, j=3$ پھر i اور j کی جگہ put کر دیں۔

$$\tan 2\theta = \frac{2a_{13}}{a_{11} - a_{33}}$$

اسی طرح اگر a_{12} ہو تو، $i=1, j=2$

$$\tan 2\theta = \frac{2a_{12}}{a_{11} - a_{22}}$$

اسی طرح اگر a_{23} ہو تو، $i=2, j=3$

$$\tan 2\theta = \frac{2a_{23}}{a_{22} - a_{33}}$$

③ فارمولہ حل کرتے ہیں θ کی ایک ویلیو آئے گی۔ اس کے لیے ایک orthogonal میٹرک بنانا ہوتا ہے۔

(Trick for Making Matrix)

اگر فارمولہ میں (a_{13}) استعمال کیا جائے تو اس کا مطلب ہے ایک اور تیسری - پہلی Row اور تیسرا کالم -

یا اسکو ایسے سمجھ لیں پہلی اور تیسری Row

Matrix کے diagonal میں ہم $\cos \theta$ آئے گا۔
 تو (a_{13}) کے حساب سے Matrix اسے بنے گا۔

$$S_1 = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

(a_{12}) ہوگا تو اس کا مطلب پہلی Row اور دوسرا کالم

$$S_1 = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(a_{23}) ہوگا تو

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

(4) Orthogonal Matrices میں put کرنے کے بعد Rotations معلوم کرنا یوں ہی ہیں مثلاً کے طور پر D_1, D_2

اور روٹیشن جب تک معلوم کرنا ہوتی ہے جب تک Matrix کی off-diagonals تمام elements / entries زبردست ہوجاؤں۔

Rotations معلوم کرنے کے لیے فارمولا :-

$$D_1 = S_1^{-1} A S_1$$

یہ Rotation معلوم کرنے کے بعد جو Matrix آئے گا پھر اس Matrix کو استعمال کر کے اگلی Rotation معلوم کرنا ہوگی۔

(5) جب آخری Rotation کے بعد جو Diagonal میں دیلے ہوئے ہوں گی وہ اس Matrix کی Eigen values ہوں گی۔

(6) Eigenvectors معلوم کرنے کے لیے جو Rotations نکالی ہوں گی ان Matrices کو آپس میں Multiply کر دیں گے اور Eigenvector آجائے گا۔

Day:

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step 4:

Put the value of two intervals $\phi(0)$ & $\phi(1)$ in

$$\phi'(x)$$

Then which value of $\phi'(x) < 1$, we let it

$$x = x_0$$

(x_0 کی وہ value لے جائے جو ϕ interval میں ہے)

Step 5: Then

ان interval سے درمیان

$$x_1 = \phi(x_0)$$

والی رقم

$$x_2 = \phi(x_1)$$

$$x_3 = \phi(x_2)$$

$$x_4 = \phi(x_4)$$